SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1741

CO1-AT1-Analytical Problem Solving

Register Number	192412093
Name	Naresh Kanna. C. M

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

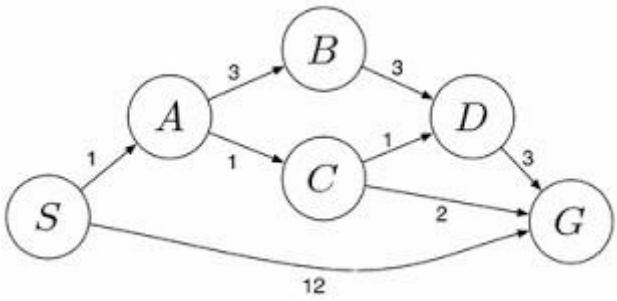
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I Naresh Kanna. C. M (192412093) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

Question 1: Water Jug Problem

Problem: You have two jugs: one with a capacity of 4 gallons and another with a capacity of 3 gallons. Neither jug has measurement markings. Obtain exactly 2 gallons in the 4-gallon jug.

Solution

Let:

- **Jug A** = 4-gallon jug
- **Jug B** = 3-gallon jug
- Initial state = (0, 0)
- Goal state = (2, 0), meaning Jug A contains 2 gallons and Jug B is empty.

Step-by-step procedure

Step	Action	Jug A (4 gallons)	Jug B (3 gallons)
1	Fill Jug B completely	0	3
2	Pour water from B into A	3	0
3	Fill Jug B completely	3	3
4	Pour from B into A until A is full	4	2
5	Empty Jug A onto the ground	0	2
6	Pour the remaining 2 gallons from B into A	2	0

Final answer

The 4-gallon jug contains exactly **2 gallons of water**, and the 3-gallon jug is empty.

AI problem formulation

1. **Initial state:** (0, 0)
2. **Goal state:** (2, 0)
3. **Actions:** Fill a jug, empty a jug, or pour water from one jug into another.
4. **State space:** All possible water quantities in both jugs, within their capacities.
5. **Solution:** Apply the above sequence of actions to reach the goal state.

Question 2: Mars Rover Intelligent Agent

A Mars Rover is an autonomous robotic vehicle that explores the surface of Mars. It collects information, analyzes soil and rock samples, and sends scientific data to Earth.

i. What are the percepts of this agent?

Percepts are the information received by the agent through its sensors.

The Mars Rover receives the following percepts:

1. **Camera images:** Images of the Martian surface, rocks, and surroundings.
2. **Distance sensors:** Information about obstacles and distances.
3. **Temperature sensors:** Temperature of the environment and equipment.
4. **Soil and rock sensors:** Information about the composition of samples.
5. **Position sensors:** Information about the Rover's location and movement.
6. **Power sensors:** Information about battery level and energy consumption.
7. **Communication sensors:** Information about communication availability with Earth.

ii. Characterize the operating environment

The Mars Rover environment can be classified as follows:

1. **Partially observable:** The Rover cannot observe the entire Martian environment at once.
2. **Stochastic:** The results of movement and exploration may be uncertain.
3. **Sequential:** Each action affects future actions and decisions.
4. **Dynamic:** Environmental conditions may change while the Rover operates.
5. **Continuous:** Position, distance, temperature, and time can vary continuously.
6. **Single-agent:** The Rover can be considered a single agent performing its exploration task, even though it may interact with other systems.

iii. What actions can the agent take?

The Mars Rover can perform the following actions:

1. Move forward and backward.
2. Turn left and right.
3. Stop when an obstacle is detected.
4. Capture images of the surface.
5. Collect soil and rock samples.

6. Analyze the collected samples.
7. Send scientific information to Earth.
8. Manage battery power.
9. Select a safe route to the target location.
10. Enter a safe mode when a fault is detected.

iv. How can we evaluate the performance of the agent?

The Rover's performance can be evaluated using the following measures:

1. **Scientific data collected:** Amount and usefulness of scientific information obtained.
2. **Sample analysis accuracy:** Accuracy of soil and rock analysis.
3. **Navigation efficiency:** Ability to reach target locations safely.
4. **Energy efficiency:** Amount of battery power consumed.
5. **Safety:** Ability to avoid obstacles and prevent equipment damage.
6. **Mission completion:** Number of assigned exploration objectives completed.
7. **Communication efficiency:** Successful transmission of useful data to Earth.

A successful Mars Rover collects accurate scientific data, completes its mission safely, and uses its resources efficiently.

v. What type of agent architecture is most suitable? Why?

A **model-based, goal-based, and utility-based intelligent agent** is suitable for the Mars Rover.

1. Model-based agent

- Maintains an internal representation of the surroundings.
- Uses previous observations to understand the environment.
- Helps the Rover navigate when the entire environment is not observable.

2. Goal-based agent

- Selects actions that help achieve specific goals.
- For example, reaching a rock formation or collecting a soil sample.
- Plans a route to reach the destination.

3. Utility-based agent

- Compares different possible actions.
- Selects actions based on safety, energy consumption, scientific value, and time.
- Helps the Rover choose the most beneficial action when several options are available.

Conclusion: A combination of model-based, goal-based, and utility-based capabilities is most suitable because the Rover must understand its surroundings, achieve scientific goals, and make efficient decisions under uncertain conditions.

Question 3: Eight Queens Problem

Problem statement

The Eight Queens Problem requires placing eight queens on an 8 × 8 chessboard so that no two queens attack each other.

A queen can attack another queen if they are in the same:

- Row
- Column
- Diagonal

Therefore, no two queens should share a row, column, or diagonal.

i. Problem formulation

1. Initial state

An empty 8 × 8 chessboard.

2. State space

All possible configurations of queens on the chessboard, including partial arrangements and complete arrangements.

3. Actions

Place a queen in an unoccupied square that does not conflict with any previously placed queen.

4. Goal test

The goal is reached when eight queens are placed on the board and no two queens attack each other.

5. Path cost

If each queen placement has a cost of 1, the total cost of placing eight queens is 8.

ii. Search strategy

A suitable method is **Backtracking Search**, which is a depth-first approach.

The algorithm places one queen in each column. If a queen conflicts with an existing queen, the algorithm tries another row. If no safe row is available, it returns to the previous column and moves that queen.

iii. Algorithm

Algorithm EightQueens(column):

```
If column > 8:
    Print the solution
    Return SUCCESS

For row = 1 to 8:

    If position (row, column) is safe:

        Place queen at (row, column)

        If EightQueens(column + 1) is SUCCESS:
            Return SUCCESS

        Remove queen from (row, column)

Return FAILURE
```

iv. Conditions for a safe position

A position (row, column) is safe if no previously placed queen has:

1. The same row.
2. The same column.
3. The same diagonal.

Since the algorithm places one queen per column, the column condition is automatically satisfied.

For queens at positions (r_1, c_1) and (r_2, c_2), a diagonal conflict occurs when:

$$|r_1-r_2|=|c_1-c_2|$$

If this condition is true, the queens attack each other diagonally.

v. One valid solution

The following row positions give a valid solution, where each number represents the row containing the queen in the corresponding column.

$$\boxed{[1,5,8,6,3,7,2,4]}$$

Column	Row containing queen
1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

Conclusion

Backtracking Search efficiently explores the search space by placing queens in safe positions and undoing placements when conflicts occur. It finds a valid arrangement in which no two queens attack each other.

Question 4: OLA Cab Booking Application

Problem statement

A customer wants to travel from a starting location to a destination using the OLA cab booking application. The customer can choose a suitable cab type, such as Mini, Micro, Sedan, Shared, or Prime.

i. Type of problem-solving agent

A **utility-based problem-solving agent** is most suitable.

Explanation

A utility-based agent evaluates different available choices and selects an option that provides the greatest benefit to the customer.

It can consider:

1. Total fare.

2. Estimated travel time.
3. Distance to the destination.
4. Customer comfort.
5. Cab availability.
6. Traffic conditions.
7. Customer preferences.

For example, if a customer wants the cheapest ride, the agent may select a low-cost cab. If the customer prioritizes comfort, the agent may select a Sedan or Prime cab.
The final choice depends on the customer's preferences and the available cabs.

ii. Pseudocode

```
Algorithm OLA_Cab_Booking

START

   Input source_location
   Input destination_location

   Input customer_preferences

   Display available cab types:
      Mini, Micro, Sedan, Shared, Prime

   Search for available cabs

   IF no cabs are available THEN
      Display "No cabs available"
      STOP
   END IF

   FOR each available cab DO

      Calculate estimated fare
      Calculate estimated travel time
      Check customer comfort preference
      Check cab availability

      Calculate utility score based on:
         Fare
         Travel time
         Comfort
         Customer preferences

   END FOR

   Select the cab with the highest utility score

   Display selected cab
   Display estimated fare
   Display estimated arrival time

   Ask customer to confirm booking

   IF customer confirms THEN
      Book the selected cab
      Display booking confirmation
   ELSE
      Cancel booking process
   END IF
```

STOP

iii. Conclusion

A utility-based agent is suitable because it compares available cab options according to cost, travel time, comfort, and customer preferences. It helps the customer choose the most appropriate cab for the journey.

Question 5: Uniform Cost Search (UCS)

Problem statement

A logistics company wants to find the minimum-cost route from starting warehouse **S** to destination warehouse **G**. Uniform Cost Search expands the node with the lowest cumulative path cost first.

The weighted graph provided in your PDF contains the following directed edges and costs. CO1-AT1-Analytical Problem Solv…

- S → A = 1
- S → G = 12
- A → B = 3
- A → C = 1
- B → D = 3
- C → D = 1
- C → G = 2
- D → G = 3

i. UCS algorithm

```
Algorithm UniformCostSearch(S, G)

   Create an empty priority queue PQ

   Insert (S, 0) into PQ

   Create an empty explored set

   WHILE PQ is not empty DO

      Remove the node N with the lowest path cost

      IF N is the goal G THEN
         Return the path and its total cost
      END IF

      IF N is not in explored set THEN

         Add N to explored set

         FOR each successor M of N DO

            Calculate:
               NewCost = Cost(N) + EdgeCost(N, M)

            Insert (M, NewCost) into PQ
            according to its cumulative cost

         END FOR

      END IF

   END WHILE

   Return FAILURE

END
```

ii. Step-by-step solution

Initially, the priority queue contains the starting node S with cost 0.

Step 1: Expand S

The available paths are:

- S → A = 1
- S → G = 12

Priority queue:

Node	Cumulative cost
A	1
G	12

Choose A because it has the lowest cost.

Step 2: Expand A

The paths through A are:

- S → A → B = 1 + 3 = 4
- S → A → C = 1 + 1 = 2

Priority queue:

Node	Cumulative cost
C	2
B	4
G	12

Choose C because its cost is 2.

Step 3: Expand C

The paths through C are:

- S → A → C → D = 1 + 1 + 1 = 3
- S → A → C → G = 1 + 1 + 2 = 4

Priority queue:

Node	Cumulative cost
D	3
B	4
G	4
G (direct path)	12

Choose D because its cost is 3.

Step 4: Expand D

The path through D to G is:

$$ S \rightarrow A \rightarrow C \rightarrow D \rightarrow G $$

Its total cost is:

$$ 1+1+1+3=6 $$

The priority queue now contains:

Node	Cumulative cost
B	4
G	4
G	6
G (direct path)	12

The lowest-cost goal path has cost 4.

Step 5: Select the goal G

The path

$$ S \rightarrow A \rightarrow C \rightarrow G $$

has total cost:

$$ 1+1+2=4 $$

UCS removes this goal node from the priority queue. Since all edge costs are non-negative, the first goal node removed from the priority queue gives the optimal path.

iii. Final answer

Optimal path:

$$\boxed{S \rightarrow A \rightarrow C \rightarrow G}$$

Minimum total cost:

$$\boxed{1+1+2=4}$$

Therefore, the least-cost route from warehouse S to warehouse G has a total cost of **4 units**.

iv. Advantages of UCS

1. Finds the optimal solution when all edge costs are non-negative.
2. Considers cumulative path cost rather than the number of edges.
3. Can find a cheaper path even when it contains more edges.
4. Is useful in route planning, transportation, and logistics problems.